

CS & IT ENGINEERING



Computer Network - 2

MAC Sub-Layer

Lecture No. - 06

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Recap of Previous Lecture



Topic

Ethernet

Topic

Back-off Algorithm



Topics to be Covered



Topic

Ethernet Standard

Topic

Ethernet Bridging

[MCQ]

[GATE-CS-2004][2 Mark]

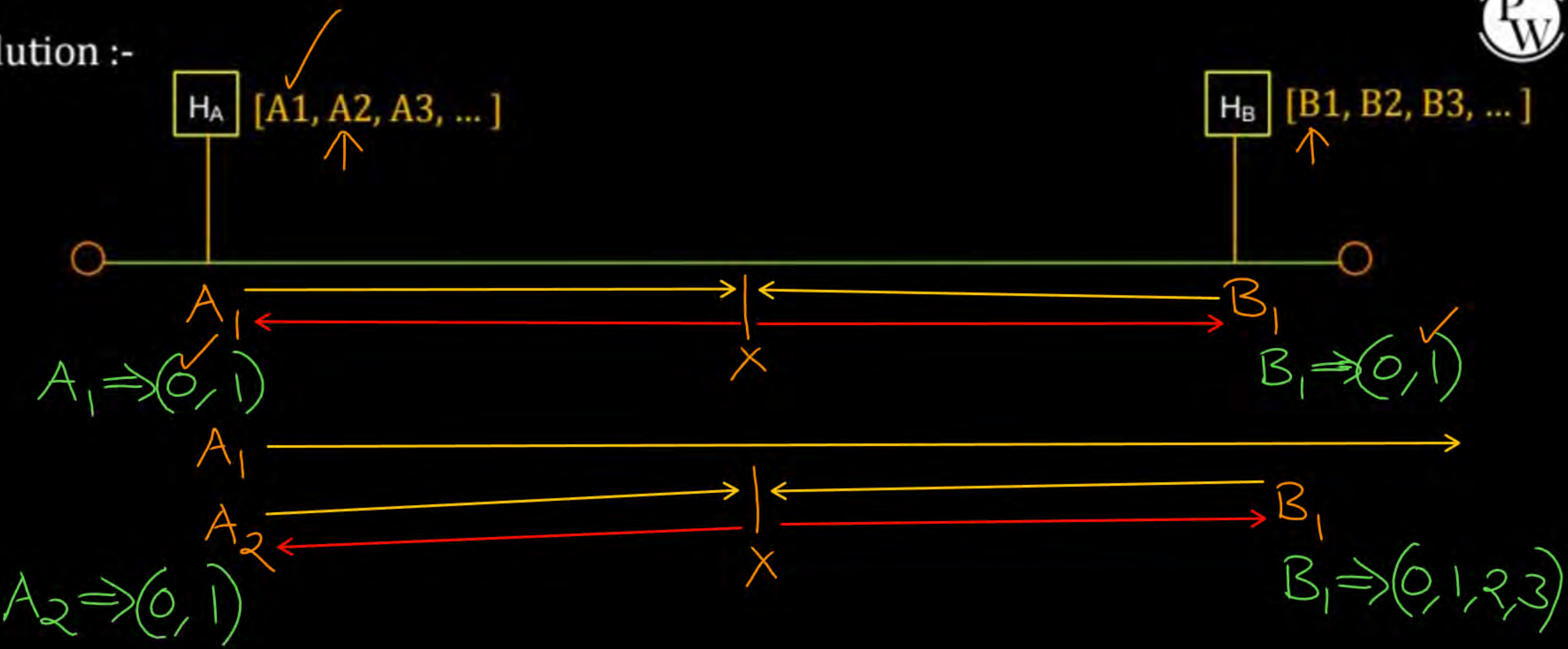


#Q. A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first back-off race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second back-off race is :

- ☐ A 0.5
- ☒ B 0.625
- ☐ C 0.75
- ☐ D 1.0

Ans: B

Solution :-



A2	B1	Result
0	0	} Again Collided
1	1	
0	1	} A win
0	2	
0	3	
1	2	
1	3	
1	0	=> B win

$$\text{Ans} = \frac{5}{8}$$



Topic : Ethernet

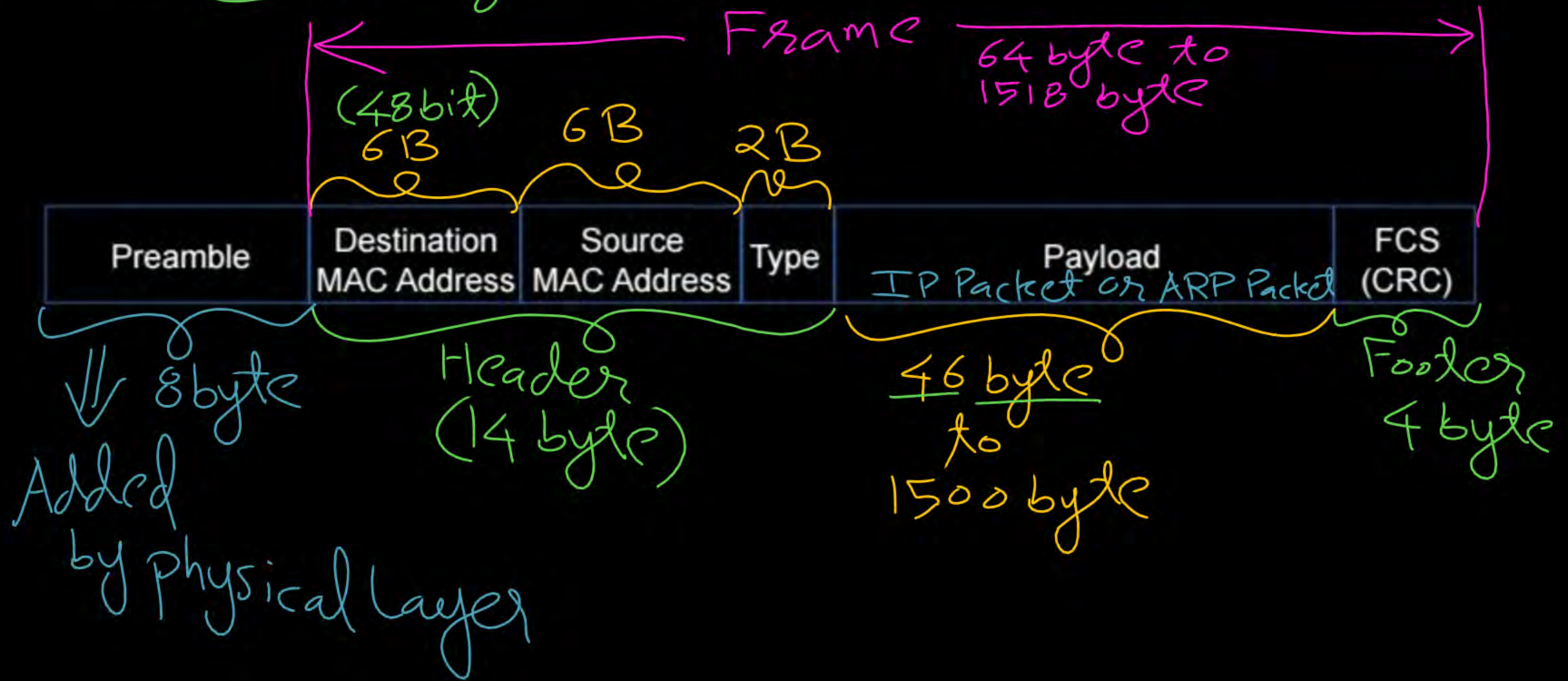


- Minimum frame size : 64 Bytes ← Based on CSMA/CD
- Maximum frame size : 1518 Bytes ← [MTU = 1500 bytes]
- Inter-frame gap between frames ✓



Topic : Ethernet Frame Format

⇒ IEEE 802.3





Topic : Preamble

$$(7+1) = 8 \text{ byte}$$



→ 7 byte binary string of alternate (1 and 0) 10101010 10

→ Allowing receivers to synchronize their clock at the bit-level with the transmitter clock.

⇒ clock synchronization

→ Preamble is followed by the one byte SFD

→ SFD ends with a 1 instead of 0

[To break the bit pattern of the preamble and signal the start of the actual frame]

SFD → 10101011

#Q. In the Ethernet, which field is actually added at the physical layer and is not part of the frame

A

preamble

B

CRC

C

address (MAC, source & Dest)

D

location (Type)

Ans: A



Topic : Type



→ Specifies the protocol of the payload
[e.g IPv4, IPv6, ARP etc]

→ if Type < 46 :
then Type is size of data in payload field bytes

→ Size of padding = [46 - Type] bytes

Type

if less than 46
→ length of
data

if more than equal
to 46

→ higher level
protocol type



Topic : FCS



→ Frame Check Sequence (FCS) $\Rightarrow$ check bit

→ 32 bits (4 bytes)

→ CRC - 32

↓
Error checking

$$G(x) = x^{32} + \dots + 1$$



Topic : Ethernet Standard



→ 10BASE

:

Ethernet

⇒ 10 Mbps

⇒ Manchester Encoding (50%)

→ 100BASE

:

Fast Ethernet

⇒ 100 Mbps

⇒ 4B/5B Encoding (80%) ✓

→ 1000BASE

:

Giga Ethernet

⇒ 1 Gbps

⇒ 8B/10B Encoding (80%) ✓

Twisted Pair:-



Full Duplex Ethernet

Ethernet

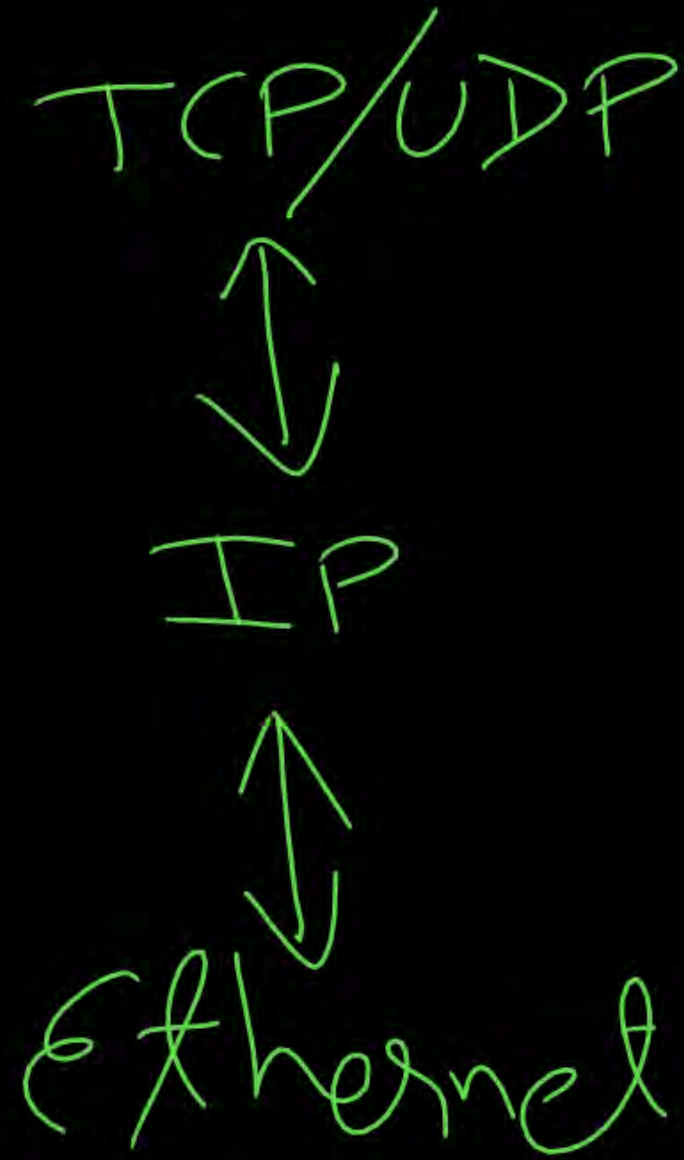
⇒ Unreliable

⇒ No Seq. no and No Ack

⇒ Pause Frame (Stop)



Half Duplex



Ethernet Efficiency:-

$$\eta = \frac{1}{1 + ka}$$

$$\left[a = \frac{\tau_P}{\tau_x} \right]$$

$$k \rightarrow 6.4, 5.4, 4.4, 3.4$$

$$\eta = \frac{1}{1 + 5a}$$



Topic : 10BASE Ethernet

→ Thick Ethernet (Thicknet) : 10 BASE 5

→ Thin Ethernet (Thinnet) : 10 BASE 2

→ 10 Mbps Bandwidth (Baseband)

→ 500 meter / 200¹⁸⁵ meter (Segment Length)

* Manchester Encoding
[Mid-bit transition]

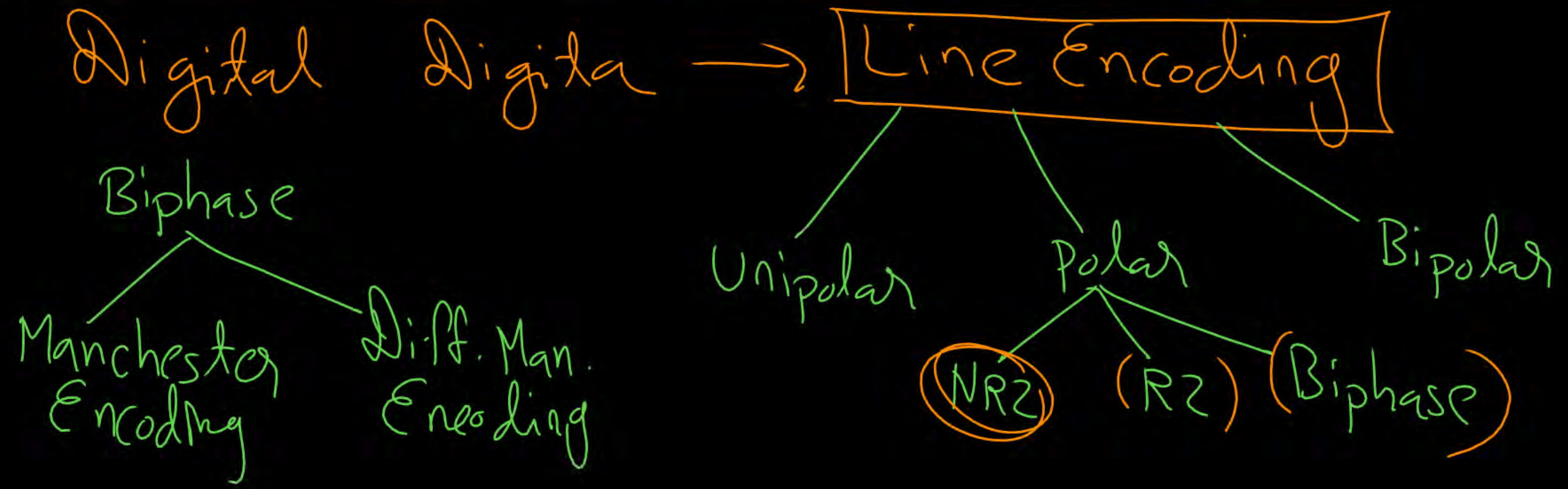
$$\text{BitRate} = \frac{\text{Baud Rate}}{2}$$

Broadband

↳ Analog
Signal



Data	Signal	
Analog	Analog	→
Analog	Digital	→
Digital	Analog	→



[MCQ]

11T-K

[GATE-CS-2007] [1 Mark]



#Q. In Ethernet when Manchester encoding is used, the bit rate is:

- ☒ A Half the baud rate. $\Rightarrow$ Two signal req. for each bit
- ☐ B Same as the baud rate.
- ☐ C Twice the baud rate.
- ☐ D None of the above

GATE-2008
Manchester
Diff. Man. Encoding

Ans: A



Topic : Ring Topology



→ Every host connected to two adjacent hosts using point-to-point link in cyclic fashion.

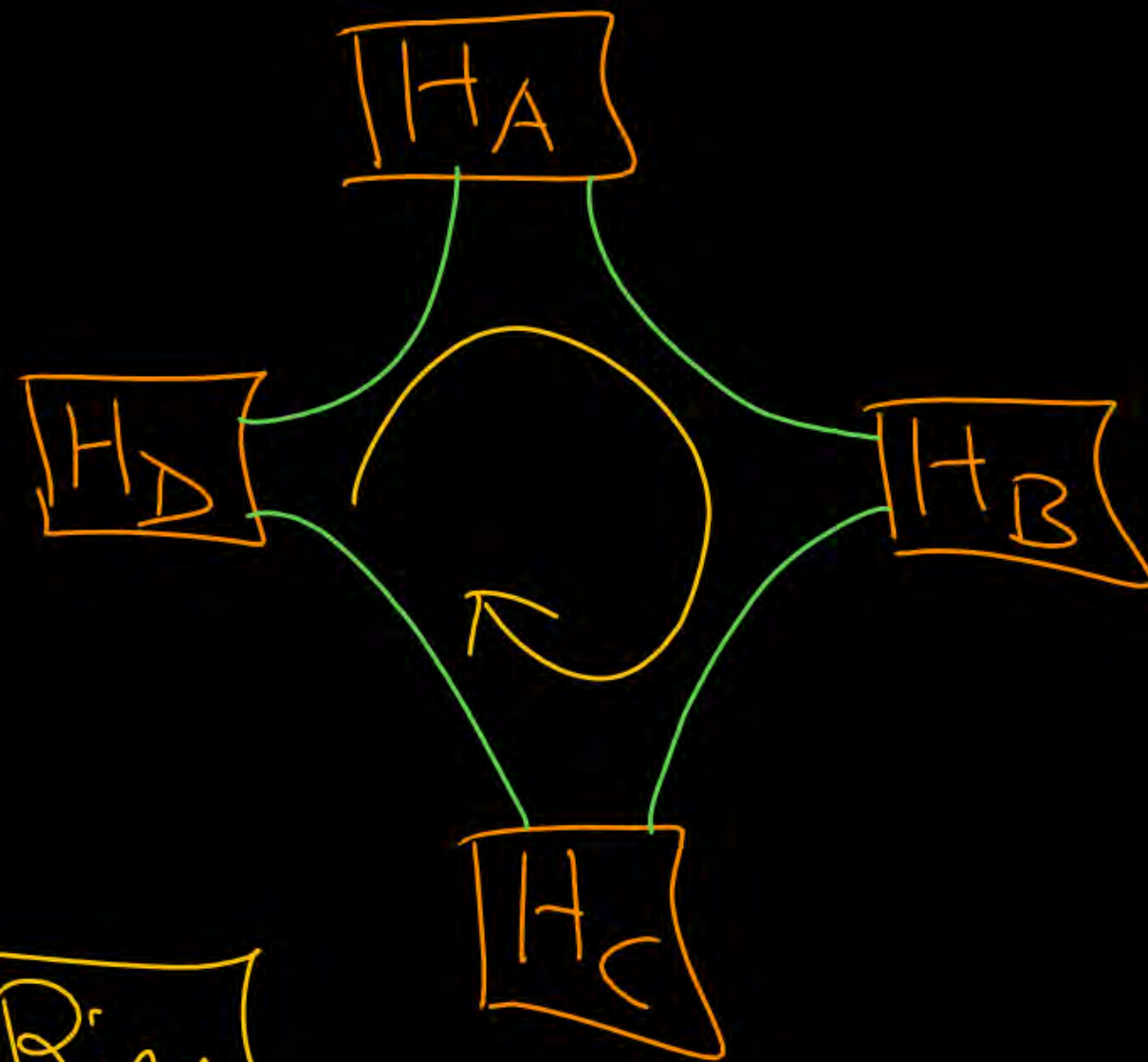
Total number of nodes = n

Total number of links = n

→ Simplex Communication

→ Access Control Method

↳ Token Passing $\Rightarrow$ Token Ring





Topic : LAN Standards

Project & IEEE 802



- IEEE 802.3 : Ethernet [Bus Topology => CSMA/CD]
- IEEE 802.4 : Token Bus [Bus Topology => Token]
- IEEE 802.5 : Token Ring [Ring Topology => Token]
- IEEE 802.11 : Wireless [Wi-Fi => CSMA/CA]

#Q. Which of the following statements is TRUE about CSMA/CD?

[GATE 2005]

H.W.

(IIT-B)

- (A) IEEE 802.11 wireless LAN runs CSMA/CD protocol.
- (B) Ethernet is not based on CSMA/CD protocol.
- (C) CSMA/CD is not suitable for a high propagation delay network like satellite network.
- (D) There is no contention in a CSMA/CD network.



Topic : Mesh Topology

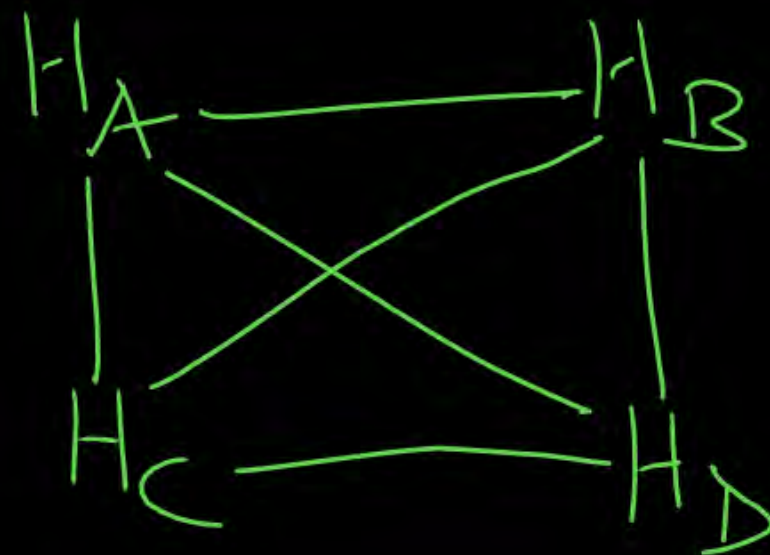


→ Every device has dedicated point-to-point link to every other device.

Total number of nodes = n

Total number of links = $\boxed{{}^nC_2}$ ← Fully Mesh Topology
$$\frac{n \times (n-1)}{2}$$
$$= O(n^2) \text{ links}$$

Internet $\Rightarrow$ Partial Mesh Topology.





Topic : Mesh Topology



Advantages :-

- Faster Communication

Disadvantages :-

- Installation cost is very high
[Preferable for small area network]
- Inefficient utilization of links
- $(n - 1)$ input-output (I/O) port per device



Topic : Star Topology



→ Every device has dedicated point-to-point link
only to a central controller.

Total number of nodes = $n + 1$

→ Hub, Switch, Router

Total number of links = n

Central Controller : Hub, Switch or Router



Topic : Star Topology



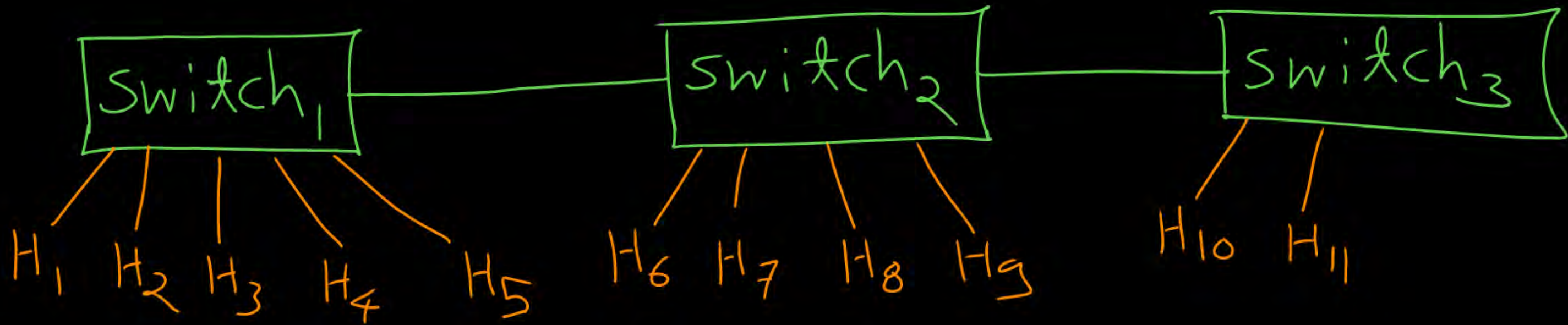
Advantages :-

- Insertion / removal of devices are easy
- Easy to extend the topology

Disadvantages :-

- If central controller fails
then all communication stop

#Q. Consider that 11 machines need to be connected in a LAN using 6-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is 3.



[NAT]

IIT-M, H.W.

[GATE-2019][2 Mark]



#Q. Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____.



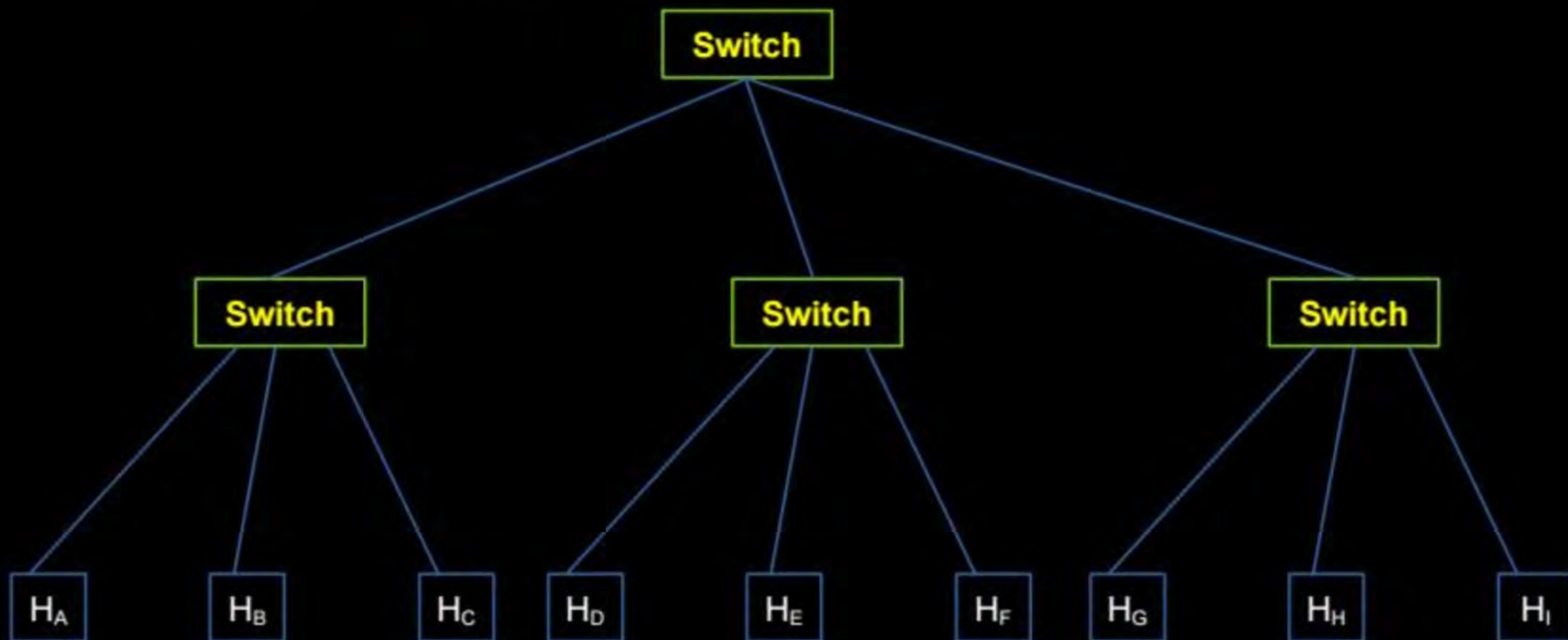
Topic : Tree Topology



- hierarchical structure
[with a central root node]
- Extension of the star topology
- Networking devices (like switches) often have "uplink" ports



Topic : Tree Topology





Topic : Uplink Port



- Special type of Ethernet port
[To connect network devices like switches or routers]
- Often designed with higher bandwidth than regular ports
[To handle larger amounts of data traffic]



Topic : Networking Devices



- Physical Layer : Layer - 1 Device
[Repeater and Hub]
- Data Link Layer : Layer - 2 Device
[Bridge / Switch]
- Network Layer : Layer - 3 Device
[Router] ✓
- Application Layer : Layer - 7 Device
[Gateway]



Topic : Repeater

⇒ Not store and Forward



→ Physical Layer [Layer - 1 Device]

→ Two port device

→ Signal Regenerator

[Regenerate the signal into output link]

→ One bit delay

[Bit by bit forwarding] ⇒ (Signal by Signal)



Topic : Repeater



End-to-end delay for one packet from Host A to Host B

$$= \underline{t_x} + \underline{t_{p1}} + \underbrace{[(1 \text{ bit}) / \text{Bandwidth}]} + \underline{t_{p2}}$$

$$= t_x + (t_{p1} + t_{p2})$$

↓
Negligible



Topic : Hub

⇒ Not store and Forward



→ Physical Layer [Layer - 1 Device]

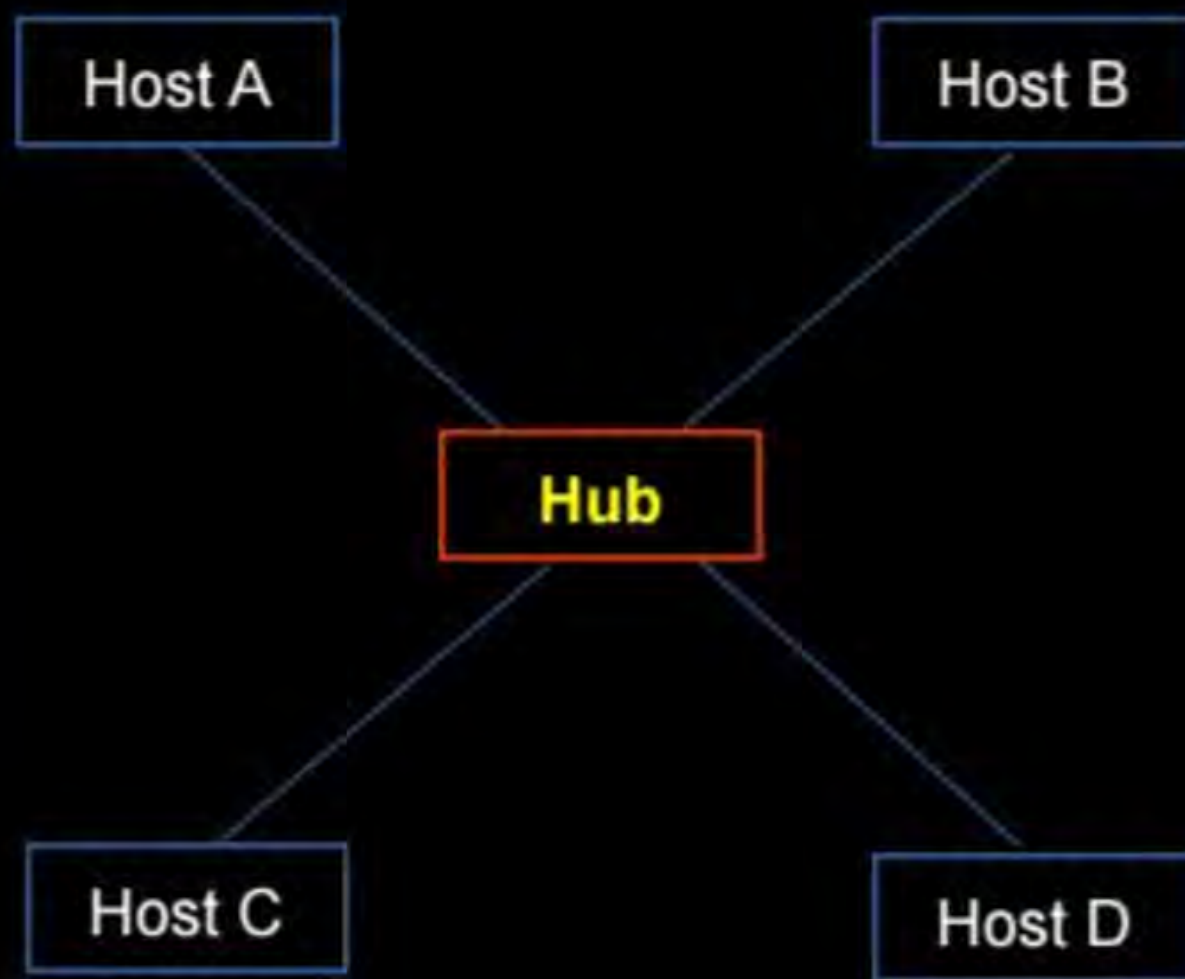
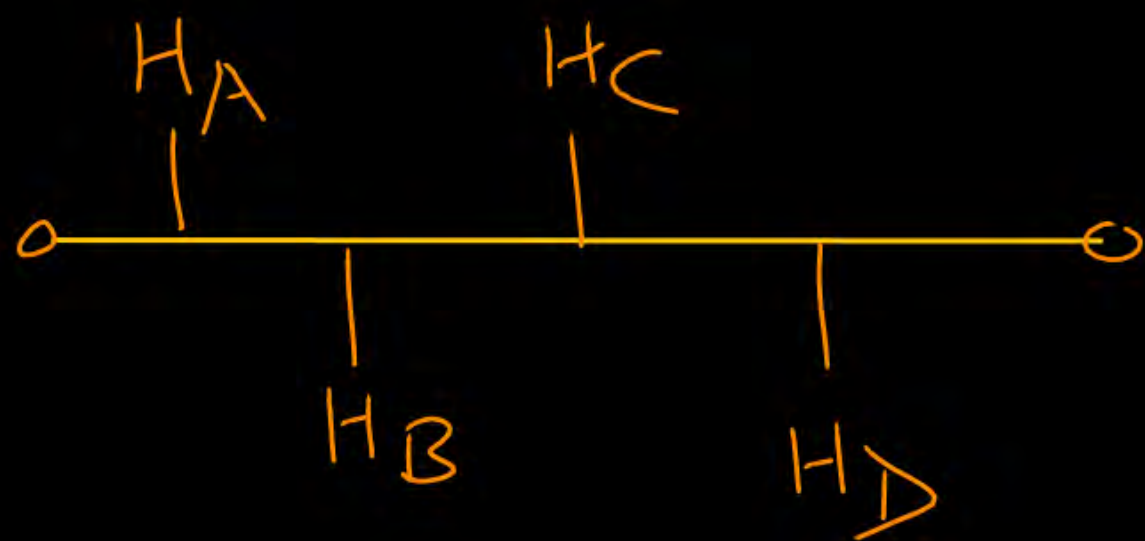
→ Multipoint repeater

→ Regenerate the signal into all connected links, except incoming link
[Broadcasting device]

→ One bit delay [Bit by bit forwarding]

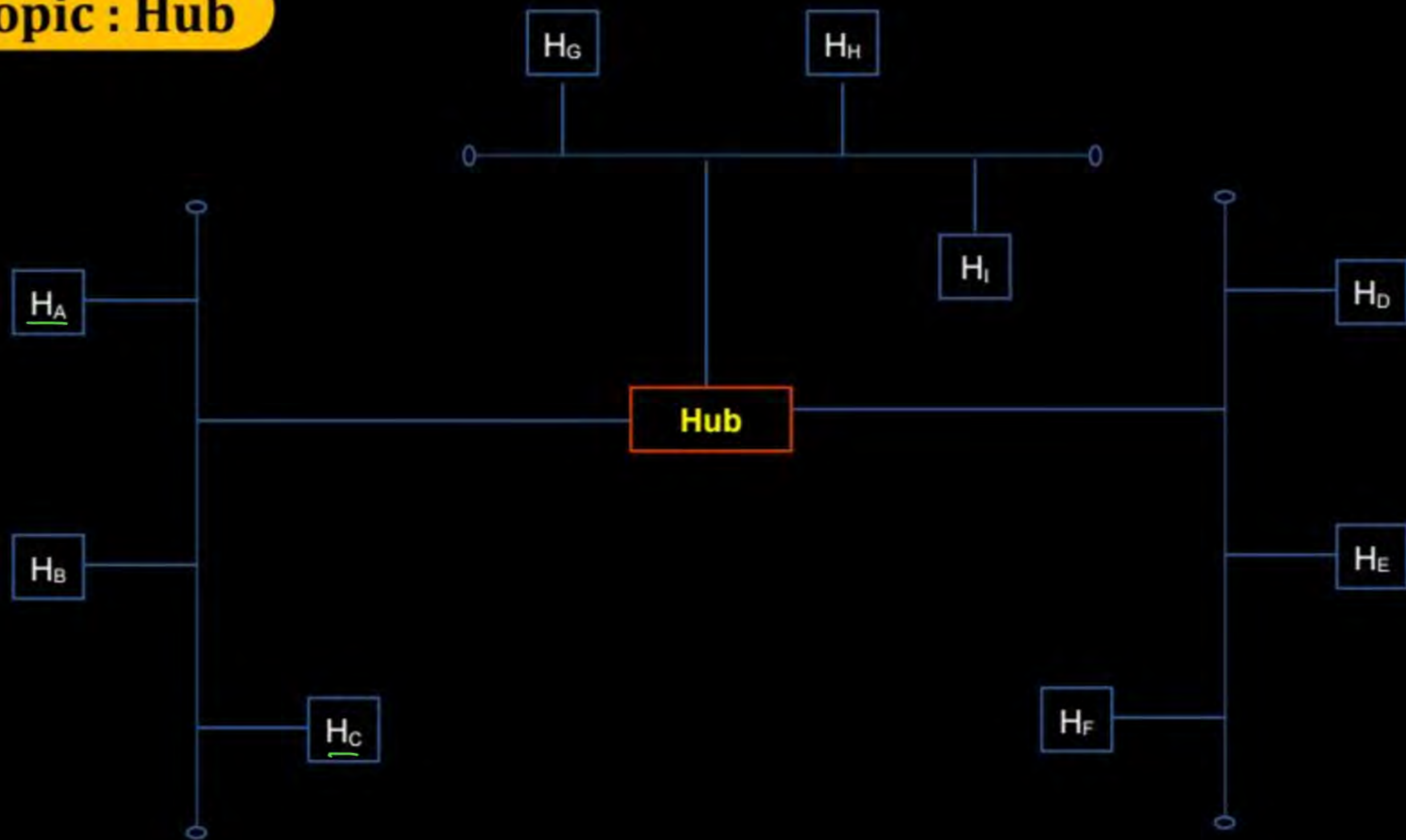


Topic : Hub





Topic : Hub



Source MAC Address = Host A MAC Address

CASE I :

Destination MAC Address = Host C MAC Address

CASE II :

Destination MAC Address = Host D MAC Address

CASE III :

Destination MAC Address = Broadcast MAC Address [FF:FF:FF:FF:FF:FF]



Topic : Hub



→ Not a broadcast domain separator
[Entire LAN has same broadcast domain]

→ Not collision domain separator
[Entire LAN has same collision domain]

Collision domain : Place where collision can occur



Topic : Switch



→ Data Link Layer [Layer - 2 Device]

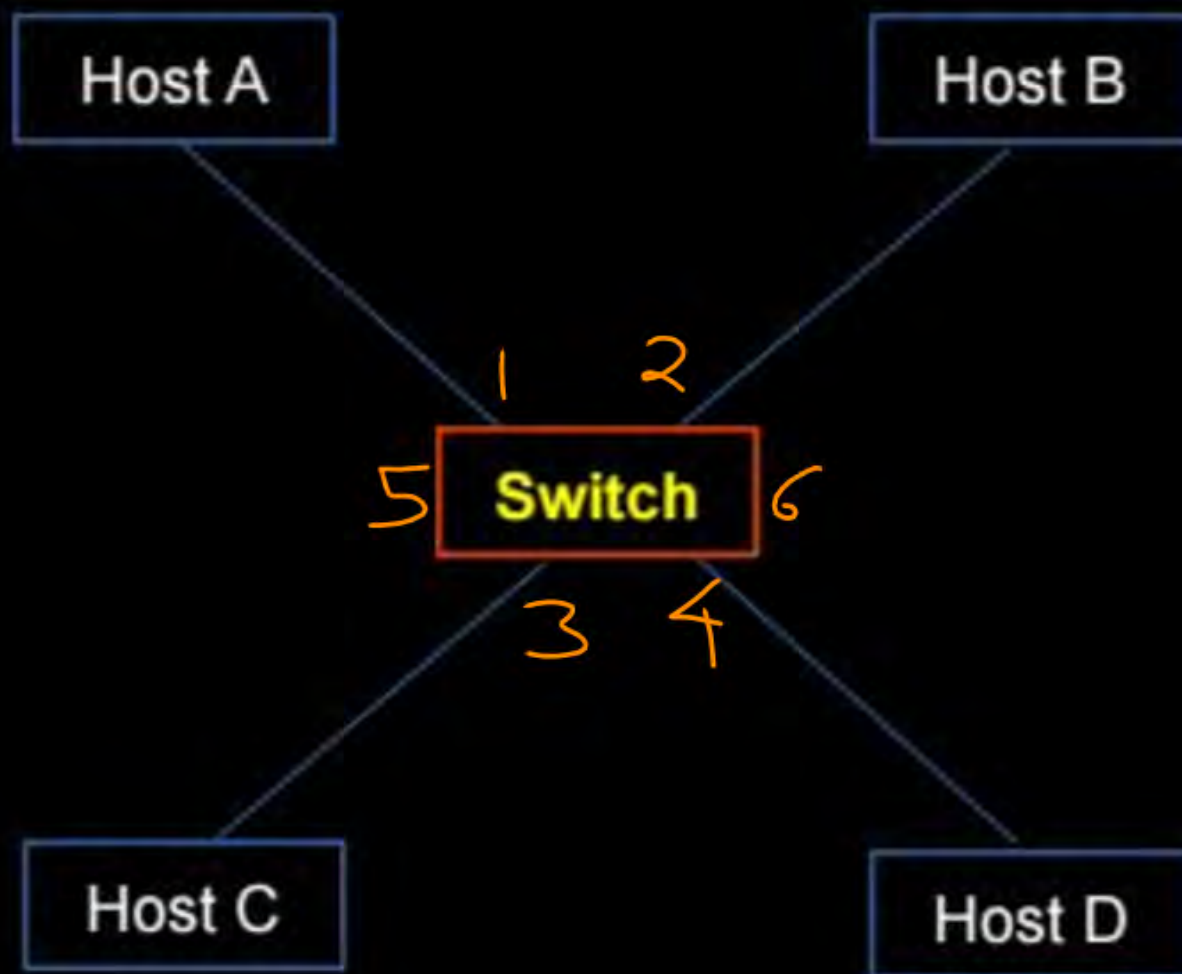
→ Old Name : Bridge
[Multiport Bridge]

→ Store and Forward
[Store, Process and Forward]

→ Switching Table [Forwarding table]
[Forwarding based on Destination MAC Address]



Topic : Switch





Topic : Switch

⇒ Transparent



Two types of switch :

1. Static

- > Non-adaptive
- > Manual update

2. Dynamic

- > Adaptive
- > Self learning
- > Plug-and-play



Topic : Switching Table

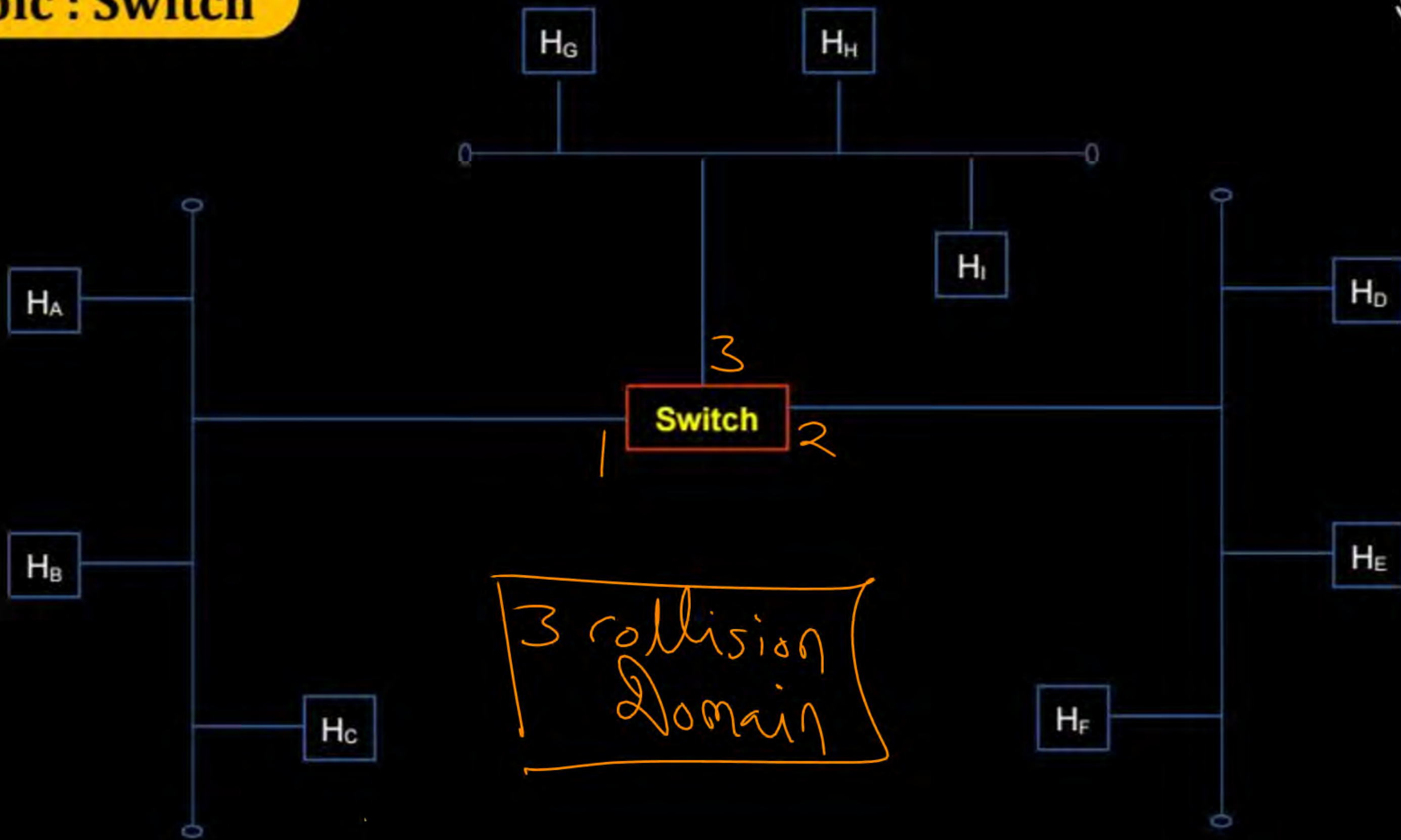


MAC Address	Interface	TTL
H _A (MAC)	1	30 sec 30
H _D (MAC)	4	30 sec
[FF:FF:FF:FF:FF:FF]	[Broadcast]	<u>Infinite</u>
[Default]	[Broadcast]	[Infinite]

Dynamic
Static



Topic : Switch



Source MAC Address = Host A MAC Address

CASE I :

Destination MAC Address = Host C MAC Address

⇒ Switch (ignore)

CASE II :

Destination MAC Address = Host D MAC Address

⇒ Switch (Forward interface)

CASE III :

Destination MAC Address = Broadcast MAC Address [FF:FF:FF:FF:FF:FF]



Topic : Switch



→ Not a broadcast domain separator
[Entire LAN has same broadcast domain]

→ Collision domain separator
[Different LAN segment has different collision domain]

Collision domain : Place where collision can occur

#Q. Which of the following statements is FALSE regarding a bridge?

- (A) Bridge is a layer 2 device
- (B) Bridge reduces collision domain
- (C) Bridge is used to connect two or more LAN segments
- (D) Bridge reduces broadcast domain

[GATE 2005]

*IIT-B
H.W.*

#Q. Ethernet layer-2 switch is a network element type which gives

H.W.

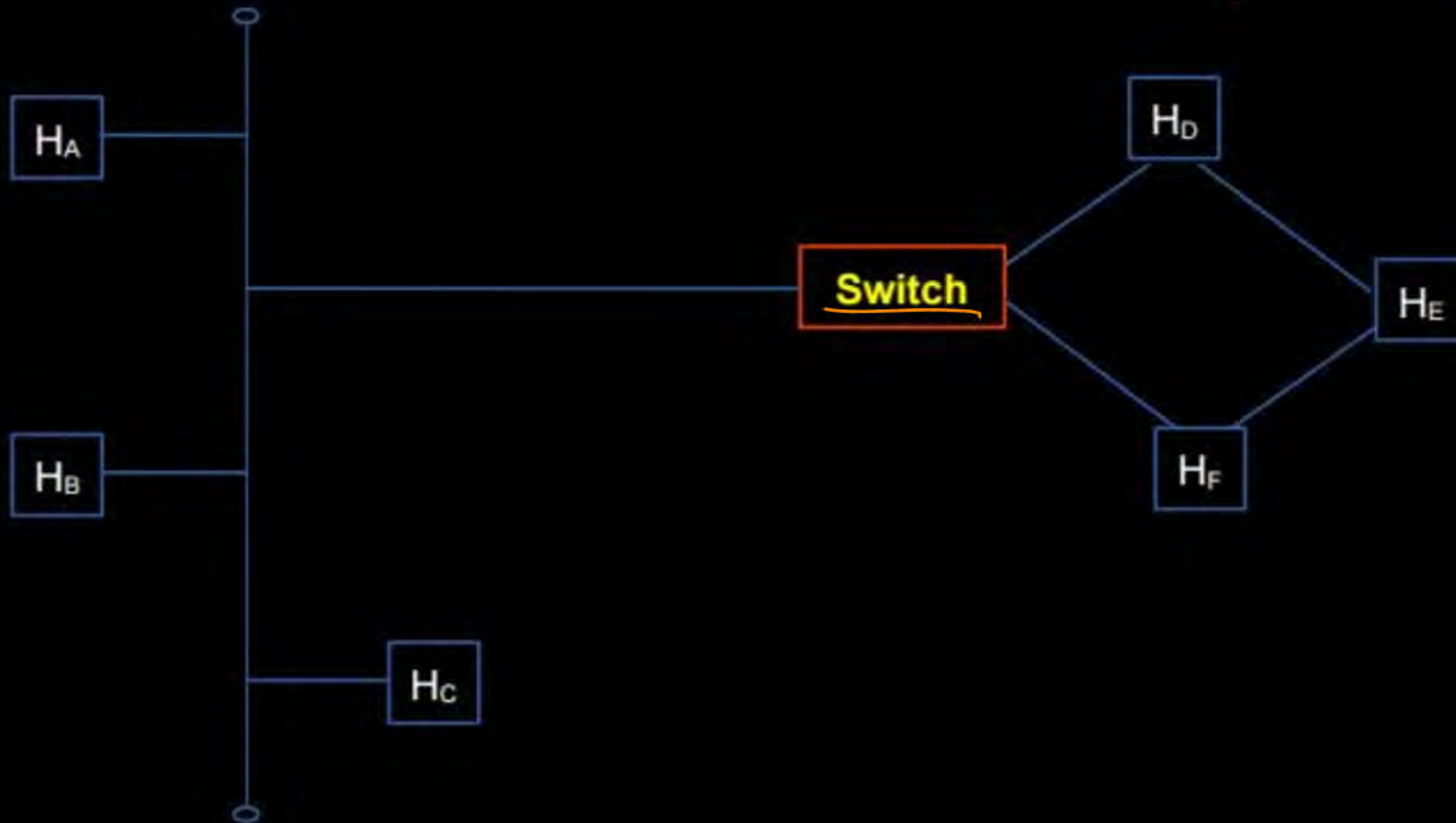
- A different collision domain and same broadcast domain
- B different collision domain and different broadcast domain
- C same collision domain and same broadcast domain
- D same collision domain and different broadcast domain



Topic : Switch



→ To connect different type of LAN segments minimum “Layer-2 device” required.





Topic : STP



- Multiple switches inside a network may form circuit (loop)
- To avoid indefinite traversing of frames switches uses "**Spanning Tree Protocol (STP)**"



#Q. In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?

- (A) For shortest path routing between LANs
- (B) For avoiding loops in the routing paths
- (C) For fault tolerance
- (D) For minimizing collisions

[GATE 2005]

IIT-B
H.W



Topic : Router



→ Broadcast domain separator
[Different subnets have different broadcast domain]

→ Collision domain separator
[Different subnets have different collision domain]

Collision domain : Place where collision can occur



2 mins Summary



Topic

Ethernet Standard

Topic

Ethernet Bridging



THANK - YOU

